

Ajay Venkatesh

Brooklyn, NY

+1 516 585 9013

av3855@nyu.edu

May 21, 2026

Hiring Manager

DentaQuest

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm finishing my M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm writing about the Business Optimization Analyst role at DentaQuest.

The role's combination of analyzing business processes, identifying optimization opportunities, and supporting data-driven decisions across operations is where my engineering background and analytical work intersect. At PwC I spent the past few years on a distributed backend on **Microsoft Azure** integrating with **Microsoft CRM** over **SQL Server**, writing complex queries, stored procedures, and parallelized batch ingestion that materially cut execution time. I built **Power BI** dashboards over datasets joined across multiple relational databases for business stakeholders.

On data preparation and operational analysis: at NYU's Novel AI Technologies I engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control, then powered **AWS QuickSight** dashboards for real-time analytics. My Big Data graduate project built a **PySpark / HDFS** pipeline over millions of flight records with rigorous EDA, anomaly detection, and feature engineering, the kind of operational-data analysis that translates into process insights.

On cross-functional collaboration: I've owned features end-to-end alongside architects, product managers, and business stakeholders at PwC, translating ambiguous operational questions into reporting and analytical deliverables for non-technical audiences. Documenting requirements and identifying process improvements is part of the consulting-style work I've already been doing.

Stack-wise: **SQL**, **Python**, **Power BI**, **Tableau**, plus **PostgreSQL**, **MySQL**, and **Snowflake** for data analysis and reporting. Comfortable with the requirements-gathering and process-mapping work that business optimization demands.

What pulls me toward DentaQuest is the operational seriousness of dental benefits administration, where careful process work compounds into real outcomes for members and providers, and the chance to use analytics to drive meaningful operational improvements.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh